

A. Identify Reusable Code

Review your existing scripts from previous assignments. Find sections of code that repeat or perform a specific, self-contained task.

B. Plan Your Functions

Note which sections can be turned into functions. Consider where **parameters**, **arguments** and **return values** could make your functions more flexible and efficient.

C. Document Your Plan

Answer the following questions in your document:

1. Describe the specific code segments you plan to modularize into functions. Explain what each function will do and whether it will have parameters, arguments, or return values.

I'm focusing on the Study Spot Finder quiz on my Welcome page. Most of the logic is inside one big click event, so I want to break that up into smaller functions.

Getting the selected answer from each radio button group

I will turn this into a function that checks which option the user picked and returns that value. This function will take one parameter (the group name) and will return the result.

Choosing the recommended study spot

I will make a function that takes the user's three answers (silence, outlets, food) and decides which study spot to recommend. This function will have multiple parameters and will return a string with the suggestion.

Showing a default message when the page loads

This will be a basic function with no parameters and no return value. It will update the page with an introductory message.

These smaller pieces make more sense as functions instead of being mixed inside the button click handler.

2. Discuss why you chose to convert these sections into functions. Focus on how modularization improves your code's readability, reduces repetition, and simplifies debugging or future updates.

I chose these sections because they handle clear, specific tasks. Modularizing the code just makes it cleaner and more organized. Breaking them into functions will:

- make the code easier to read, since each part has a name and a purpose
- reduce repeated code, especially for checking radio buttons
- make debugging easier, because each piece is separate and easier to test
- make future changes simpler, since I can update one function without touching the whole script

3. Propose any new functions that could enhance your site's features or improve user interaction. For each new function, describe its purpose, what parameters it will take, whether it will return a value, and how it fits into your overall project.

I might add a couple of small helper functions later:

1. **A function to show a message** (like an error or success message). It would take the element ID and the text I want to show. No return value needed.
2. **A function to clear a form** after the user submits something. It would take the form ID as a parameter and simply reset it.

These would make my site easier to manage and would reduce repeated lines of code.